

OC Core 1.3 / Physics Domain Packet

Status Snapshot

- Current state: `VALIDATED_ANCHOR_ACTIVE`
- Target state: `VALIDATED_ANCHOR_ACTIVE`
- Scientific class: `THEOREM_NATIVE`
- Trace status: `TRACE_COMPLETE`
- Closure verdict: `PASS`
- Quantitative pass result: `PASS`
- Next required action: `MAINTENANCE_ONLY`
- Replay status: `PASS_REPLAYABLE`

Theorem-to-Observable Map

- K1/K2 contradiction-load and stabilization-margin operators lawfully generate the bounded physics packet over constants, Balmer spectra, and transport-scaling families.
- Root-law thresholds are exposed as theorem-native residual observables on the pinned official routes rather than as unrestricted physical universalization claims.
- Every promoted benchmark family is traced through the locked K1/K2/K9/K10 route, official datasets, and held-out replay.

Benchmark Dataset Manifest

- `PHY_BENCH_001`: `fine_structure_constant_residual` | NIST / CODATA constants | <https://physics.nist.gov/cuu/Constants/index.html> | held-out: leave-one-family-out residual verification
- `PHY_BENCH_002`: `hydrogen_balmer_line_residual` | NIST Atomic Spectra Database | https://physics.nist.gov/PhysRefData/ASD/lines_form.html | held-out: reserve transitions not used during packet fitting
- `PHY_BENCH_003`: `transport_scaling_residual` | DOE Office of Science challenge corpora | https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_Grand_Challenges_Report_final.pdf | held-out: hold out one transport regime family

Pinned Official Route Protocol

- `PHY_ROUTE_001`: NIST (`OFFICIAL_CONSTANTS_PORTAL`) -> <https://physics.nist.gov/cuu/Constants/> | purpose: Pinned constants and CODATA reference values
- `PHY_ROUTE_002`: NIST (`OFFICIAL_SPECTRA_DATABASE`) -> <https://physics.nist.gov/PhysRefData/> | purpose: Pinned spectral benchmark families
- `PHY_ROUTE_003`: U.S. Department of Energy Office of Science (`OFFICIAL_CHALLENGE_PROGRAM`) -> <https://science.osti.gov/ascr> | purpose: Transport or plasma-scale challenge framing and official program context

Measurable Outputs

- `relative_constant_error`
- `spectral_line_residual`
- `transport_scaling_residual`

Measurement Schema

Bind theorem-to-observable maps to pinned official constants and spectra, then evaluate held-out residual envelopes.

Acceptance Criterion

Held-out residuals stay inside the declared tolerance band for every promoted benchmark family.

Falsifier

Any benchmark family with residuals outside tolerance or with broken sign/order constraints falsifies the promoted packet.

Quantitative Acceptance Thresholds

- coverage_ratio_required: 1.0
- critical_residual_sigma_threshold: 2.5
- normalized_error_max_max_sigma: 5.0
- normalized_error_mean_abs_max_sigma: 1.0
- normalized_error_p95_max_sigma: 2.5
- policy_snapshot: {'benchmark_residual_policy': {'critical_residual_sigma_threshold': 2.5, 'schema_id': 'LOGION_BENCHMARK_RESIDUAL_POLICY_v1', 'severe_residual_sigma_threshold': 1.5}, 'prediction_accuracy_policy': {'brier_score_max': 0.08, 'cases_total_required': 30, 'coverage_ratio_required': 1.0, 'critical_failure_f1_min': 0.87, 'critical_failure_precision_min': 0.85, 'critical_failure_recall_min': 0.9, 'expected_calibration_error_max': 0.05, 'schema_id': 'LOGION_PREDICTION_ACCURACY_POLICY_v2'}, 'prediction_sigma_policy': {'cases_total_min': 30, 'normalized_error_max_max_sigma': 5.0, 'normalized_error_mean_abs_max_sigma': 1.0, 'normalized_error_p95_max_sigma': 2.5, 'schema_id': 'LOGION_PREDICTION_SIGMA_POLICY_v1', 'tail_breach_count_max': 0}}
- severe_residual_sigma_threshold: 1.5
- tail_breach_count_max: 0

Replay Harness

- Replay command: `validation/run_all.py --qa-only` (public replay umbrella; domain: PHYSICS)
- Evidence bar: HYBRID_ESCALATION
- Protocol id: OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT

Institute-run Escalation

- Current escalation status: NOT_REQUIRED_UNLESS_NEW_OBSERVABLE_FAMILY_ENTERS
- Escalation trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSONALITY
- Measurement wave id: INSTITUTE_RUN::PHYSICS::WAVE_1A
- Measurement plan: Acquire partner or institute-run measurements for the missing observable family and rerun the held-out replay packet without widening scope.

Blocking IDs

- None.